

# ENZYMES

Enzymes are proteinaceous substances or compounds which can catalyze chemical reactions of biological origins without themselves undergoing any changes. So, they are called biocatalyst.

An enzyme may be defined as a protein that enhances the rate of biochemical reactions but does not affect the nature of final product.

The molecules upon which enzymes may act are called substrate and the enzyme converts the substrates into different molecules known as products.

The study of composition and function of enzyme is known as enzymology.

# Characteristics (properties of enzymes)

Enzymes show following characters

1. Protein nature: Enzymes are generally globular proteinaceous structure.
2. Molecular weight: Enzymes are large macromolecules.
3. Chemical reaction: Enzymes may increase the rate of chemical reaction.
4. Reversibility: all enzymes-controlled reactions are reversible.
5. Enzyme specificity: Enzymes are highly specific in their action.
6. pH: Each enzyme functions at a particular pH but not in variable pH.
7. Heat sensitive: All enzymes are heat sensitive.
8. Solubility: Majority of enzymes are water soluble, and some enzymes are insoluble in water.

# Classification of enzymes

## Six main classes of enzymes

- **Oxidoreductases (EC 1):** Catalyze oxidation-reduction reactions, such as the transfer of electrons or hydrogen atoms. [ⓘ](#)
- **Transferases (EC 2):** Transfer a functional group from one molecule to another (e.g., methyl group, phosphate group, or amino group). [ⓘ](#)
- **Hydrolases (EC 3):** Catalyze the cleavage of a bond by the addition of water (hydrolysis). [ⓘ](#)
- **Lyases (EC 4):** Break chemical bonds (like C-C, C-N, or C-O) without hydrolysis and without oxidation. They can also catalyze the formation of double bonds. [ⓘ](#)
- **Isomerases (EC 5):** Catalyze geometric or structural changes within a single molecule rearranging atoms. [ⓘ](#)
- **Ligases (EC 6):** Join two molecules together by forming a new covalent bond, often using energy from ATP hydrolysis. [ⓘ](#)

## Seventh class

- **Translocases (EC 7):** Catalyze the movement of substances across a biological membrane. [ⓘ](#)

# Functions (biological roles)

1. ATPase helps in the formation of energy in the form of ATP by the process of phosphorylation.
2. Enzymes are controlled by specific genes. Activation or repression of genes allow certain enzymes to be functional or nonfunctional in cells respectively.
3. Transcription , translation and DNA replication are controlled by RNA polymerase, peptidyl transferase and DNA polymerase respectively.
4. Oxidoreductase catalyzes the oxidation reaction where the electrons trend to travel from one form of a molecule to the other.

5. Transferase help in the transportation of the functional group among acceptors and donors molecules.
6. Hydrolases catalyze the hydrolysis reaction by adding water to cleave the bond and hydrolyze it.
7. Isomerase catalyze the structural shifts present in a molecule, thus causing the change in the shape of the molecule.
8. Ligases catalyze the joining of two molecules. For example, DNA ligase catalyzes the joining of two fragments of DNA by forming a phosphodiester bond.
9. Enzymes are used in chemical industry and other industrial applications when extremely specific catalysts are required.